Class-4 Cellular Automata as Reservoirs and Programmable Substrates

A study of fractal rule generation, dimensional transfer, and the limits of CA reservoir computing

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Abstract

We investigate networks of class-4 (“edge of chaos”) hexagonal cellular automata (CAs), with K=4 states and a 7-cell pointy-top neighbourhood, as reservoirs for byte-level sequence prediction and as substrates for conditional computation. Rules are generated by an unusual method: posterising Mandelbrot/Julia/Newton escape-time images into K=4 transition tables. Across three corpora (1920–40 news, source code, multilingual text) and eight random seeds, an evolved class-4 reservoir beats a random-rule reservoir decisively (Cohen’s *d* = 2.2–3.9). However, head-to-head, class-4 rules merely **tie** structured *linear* (class-3) rules — so the operative principle is *structured* ≫ *random*, not that class-4 is uniquely privileged, in agreement with the reservoir-computing literature. The most distinctive results are on the generator side: fractal-derived rules are class-4 at a **dimension-invariant ≈⅓ rate** for lattice dimension $N = 2\ldots5$ (random rules: ≈0 % at $N \geq 2$), and a 2D-hexagonal class-4 rule remains class-4 in 91 % of cases when re-interpreted on a 3D cubic lattice. For prediction, the binding constraint is the linear readout, not the reservoir: scaling, depth, symmetry priors, extra input ports, and data-conditioned rule selection all plateau at the same ceiling. We also demonstrate a programmable CA (“cell-11”) supporting instruction-driven operations, glider signal transport, and a port-controlled 1-bit gate; glider-collision logic gates are structurally precluded because fractal-derived rules are anisotropic.

1. Introduction

Reservoir computing exploits a fixed, untrained nonlinear dynamical system to project inputs into a high-dimensional feature space from which a cheap linear readout learns. Cellular automata are an attractive reservoir substrate: discrete, GPU-free, fully inspectable. The “edge of chaos” hypothesis suggests class-4 CAs — those exhibiting persistent, mobile, localised structures — should make the best reservoirs. We test this with a hexagonal, four-state CA whose rules are generated by a fractal procedure, and report a calibrated account of what works, what does not, and what is novel.

2. Background and prior art

Reservoir computing with CAs (ReCA) was introduced by Yilmaz (2014–2015) and named by Margem & Yilmaz (2016), with non-uniform and coupled extensions by Nichele and colleagues (2017). **Genetic search over CA rules** for computation is older still (Packard 1988; Mitchell, Crutchfield & Hraber 1993). **Hexagonal, multi-state CAs** — including explicit four-state hexagonal examples and input-entropy classification of their complex dynamics — are well covered by Wuensche & Gómez-Soto’s DDLab framework, and hexagonal multi-state computation appears in Wuensche & Adamatzky’s “Spiral rule”. The **“edge of chaos is optimal”**

claim is contested: the best ReCA rules are often class-3 (Yilmaz 2015); Langton's λ alone does not determine behaviour class (Sakai & Kanno 2002); and Carroll (2020) questions whether reservoirs work best at the edge of chaos. The element with **no located prior art** is the generation of CA rule tables from *escape-time* fractals (Mandelbrot/Julia), and the use of CAs with extra routable input ports beyond the local neighbourhood.

3. Methods

Substrate. A K=4 hexagonal CA on a toroidal lattice; the local rule maps the 7-cell neighbourhood (self + 6 neighbours) to a next state via a 4^7 = 16,384-entry lookup table. Stepping is vectorised and verified bit-identical to a reference implementation.

Fractal rule generation. An escape-time fractal is rendered, its escape grid posterised into K=4 levels, and the flattened image used as the rule table. Candidate rules are screened for class 4 via tail-activity of a noisy seed (the band of "alive but not chaotic" activity).

Reservoirs and search. Multiple CA nodes form a directed graph; a genetic algorithm evolves node count, board size, tick count, coupling, rule choice, and topology. A held-out linear/logistic readout is trained on reservoir features concatenated with a short byte context; fitness is held-out accuracy with a matched random-rule control. Larger searches use island models and hierarchical ("deep") reservoirs.

Extra-input CAs (cell8/10/11). The neighbourhood is extended with 1, 3, or 4 extra routable input ports; the rule table is indexed by the neighbourhood plus the ports, with the all-ports-zero slice equal to the base rule (ports inert by default).

Compute. Embarrassingly parallel screens (rule libraries, dimension surveys, glider/collision searches) ran on the ALICE HPC cluster (Leiden) via SLURM array jobs.

4. Results

4.1 Structured reservoirs beat random — robustly

corpus	class-4 acc.	random acc.	Cohen's *d*
news	0.199 ± 0.039	0.127 ± 0.027	2.2
code	0.101 ± 0.023	0.033 ± 0.007	3.9
languages	0.177 ± 0.027	0.096 ± 0.020	3.4

Held-out next-byte accuracy, eight seeds per cell. The class-4 reservoir's advantage over a random-rule reservoir is large and consistent.

4.2 …but class-4 is not uniquely special

Adding a *linear* (additive mod-4, class-3-typical) rule arm: class-4 **ties** linear (news d = +0.2, code +0.6, languages 0.0); both far exceed random. The operative principle is *structured* ≫ *random*, consistent with Yilmaz (2015) and Carroll (2020).

4.3 Fractal family and class-4 yield

generator	class-4 yield	note
Julia	45 %	highest yield
Newton	35 %	most diverse rules
Burning Ship	30 %	
Mandelbrot	25 %	
Multibrot (z^3)	17 %	tends chaotic

4.4 Dimension-invariant class-4 yield, and 3D transfer

Re-interpreting the rule table on an N-dimensional von Neumann lattice, the fraction of fractal-derived rules that are class-4 stays near one-third for $N = 2\ldots5$ (32.5 / 33.6 / 31.1 / 31.5 %), whereas random rules collapse to ≈0 % at $N \geq 2$. A 2D-hexagonal class-4 rule remains class-4 in **91 %** of cases when run as a 3D cubic rule (54,191-rule library).

4.5 Symmetry priors

Enforcing hexagonal rotation (C6) or rotation+reflection (D6) symmetry shrinks the rule space 5.9× / 9.5× but lowers class-4 yield (43 % → 26–28 %; enforced isotropy over-stabilises). Symmetry is therefore best used as a soft, search-selectable prior.

4.6 Reservoir vs n-gram, and the mixture

A standalone class-4 reservoir reaches ≈3.78 bits/byte versus ≈3.51 for a plain n-gram. An arithmetic mixture (0.88 n-gram + 0.12 reservoir) reaches 3.771 — a small (+0.037 bits/byte) but interior-weighted improvement, indicating the reservoir carries a little information the n-gram lacks.

4.7 cell-11 as a programmable CA

With ports treated as instruction opcodes, the substrate executes instruction streams exactly (RUN/INC/SHIFT/CLEAR), composes them over ticks, and reverts to base class-4 dynamics when ports are quiet. Among quiescent-background rules, gliders are abundant (≈37 % support a moving localised structure). A port-written **sink barrier** cleanly blocks a glider signal, realising a 1-bit gate under program control. Two-glider **collision** logic gates, however, are unreachable: across 2,500 rules, none produced a *convergent* glider pair — fractal posterisation breaks rotational symmetry, so gliders are unidirectional and cannot collide.

5. The readout ceiling (negative results)

Every reservoir-side lever plateaued at the same point: island-model scaling with depth-3 hierarchical reservoirs converged to small networks; extra input ports (cell8/10/11) failed to help and were discarded by the search; data-conditioned (matching-pursuit) rule selection tied a random draw on held-out data. The binding constraint is the span of the linear readout over the reservoir features, not the reservoir's size or construction.

6. Novelty assessment

Re-derived prior art: ReCA, GA-over-CA, coupled-CA reservoirs, and the hexagonal K=4 class-4 substrate itself (Wuensche/DDLab). Genuinely unexplored, as far as a targeted literature search could determine: (i) escape-time-fractal → CA-rule generation; (ii) the dimension-invariance of class-4 yield for such rules; (iii) CAs with extra routable input ports as a conditional/programmable interface. Absence of evidence is not proof of novelty; Wuensche's full DDLab corpus and informal sources were not exhaustively searched.

7. Limitations

Results are at modest corpus scale; the readout is a simple linear/logistic layer; the n-gram comparison uses an interpolated byte model; novelty claims rest on a non- exhaustive search; and the programmable-computation results are primitives, not a full logic family.

8. Conclusion

Class-4 hexagonal CAs make competent reservoirs — decisively better than random rules, but no better than other structured rules — and the linear readout, not the reservoir, sets the prediction ceiling. The distinctive contributions are the fractal escape-time rule generator and its dimension-invariant class-4 yield. As a programmable substrate, the extended-port CA supports instruction-driven computation and gated signal transport, while collision-based logic awaits isotropic rule families.

References

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